

Lecture Unit 2.2

OpenMP “Hello World” Demo

About the demo

- Will build and run the OpenMP “Hello World” program from the previous unit on a compute node in ISCA
- The build will use the GCC C compiler (<https://gcc.gnu.org>)
- Will show how to get the number of processors available to the operating system by looking in `/proc/cpuinfo`
- Will show how to control the number of threads using the `OMP_NUM_THREADS` environment variable

Demo

[illegible]

Demo

For user documentation please visit the "Advanced Research Computing Wiki" via
<https://universityofexeteruk.sharepoint.com/sites/ExeterARC>

Send any direct support issues ISCA / HPC via
<https://uoeitservicedesk-apps.easyvista.com/s/selfservice>

Please DO NOT run tasks on ISCA's head, login or compute nodes outside of the Slurm workload management system! Such jobs may be killed without notice due to the negative impact on all cluster users.

SLURM: Your account, jp1143, is ready to submit Slurm jobs.
Info: Loaded slurm/23.02.6 into the modular environment.
Info: Loaded lmod into the modular environment.

```
(OSS 2407.0) [jp1143@login02(exeter) ~]$ ls
HPC  TeachingW2  Testing  Week2WS
(OSS 2407.0) [jp1143@login02(exeter) ~]$ cd TeachingW2
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ ls
calc_sum.c  calc_sum_reduction.c  hello.c  hello.o  loop2.c  loop3.c  loop.c
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ |
```

Demo

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calc_sum.c calc_sum_reduction.c hello.c hello.o loop2.c loop3.c loop.c  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ nano hello.c
```

Demo

```

/*****
Description: The Hello World program from ECM3446/ECMM461 unit 2.2

Purpose: demonstrate the use of an OpenMP pragma
        by printing "Hello World!" from each thread

Notes: compile with the appropriate OpenMP flag for
       your compiler e.g. gcc -fopenmp -o hello hello.c

*****/

#include <stdio.h>

int main(){
#pragma omp parallel
{
    printf("Hello World!\n");
}
    return 0;
}

```

Demo

```
-----  
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calc_sum.c calc_sum_reduction.c hello.c hello.o loop2.c loop3.c loop.c  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ gcc -o hello hello.c  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ ls  
calc_sum.c calc_sum_reduction.c hello hello.c hello.o loop2.c loop3.c  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ |
```


Demo

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(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ gcc -o hello hello.c  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ ls  
calc_sum.c calc_sum_reduction.c hello hello.c hello.o loop2.c loop3.c  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ ./hello  
Hello World!  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ |
```


Demo

[illegible]

Demo

```
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ cat /proc/cpuinfo|
```

Demo

```
processor      : 47
vendor_id     : GenuineIntel
cpu family    : 6
model         : 106
model name    : Intel(R) Xeon(R) Gold 5317 CPU @ 3.00GHz
stepping      : 6
microcode     : 0xd0003d1
cpu MHz       : 3000.000
cache size    : 18432 KB
physical id   : 1
siblings      : 24
core id       : 11
cpu cores     : 12
apicid        : 87
initial apicid : 87
fpu           : yes
fpu_exception : yes
cpuid level   : 27
wp            : yes
flags         : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm
               pbe syscall nx pdpe1gb rdtscp lm constant_tsc art arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf pni pcl
mulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid dca sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_ti
mer aes xsave avx f16c rdrand lahf_lm abm 3dnowprefetch cpuid_fault epb cat_l3 intel_ppin ssbd mba ibrs ibpb stibp ibrs_enhanced tpr_
shadow flexpriority ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid cqm rdt_a avx512f avx512dq rdseed adx smap a
vx512ifma clflushopt clwb intel_pt avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves cqm_llc cqm_occup_llc cqm_mbm_tot
al cqm_mbm_local split_lock_detect wbnoinvd dtherm ida arat pln pts hwp_epp vnmi avx512vbmi umip pku ospke avx512_vbmi2 gfni vaes vpc
lmulqdq avx512_vnni avx512_bitalg tme avx512_vpopcntdq la57 rdpid fsrm md_clear pconfig flush_l1d arch_capabilities
vmx flags     : vnmi preemption_timer posted_intr invvpid ept_x_only ept_ad ept_lgb flexpriority apicv tsc_offset vtptr mtf vapid ep
t vpid unrestricted_guest vapid_reg vid ple shadow_vmcs pml ept_mode_based_exec tsc_scaling
bugs          : spectre_v1 spectre_v2 spec_store_bypass swaps mmio_stale_data eibrs_pbrsb gds
bogomips      : 6000.00
```

Demo

```
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ export OMP_NUM_THREADS=4  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$ ./hello  
Hello World!  
Hello World!  
Hello World!  
Hello World!  
(OSS 2407.0) [jp1143@login02(exeter) TeachingW2]$
```

Compiler flags

- We need to give the compiler a flag to tell it to use OpenMP directives to parallelise the executable
- Compiler flags are not specified in the standard and differ between compilers e.g.
 - GNU C compiler: `gcc -fopenmp`
 - Intel C compiler: `icc -qopenmp`

The OMP_NUM_THREADS environment variable

- We can use the **OMP_NUM_THREADS** environment variable to control the number of threads forked in a parallel region e.g.
 - `export OMP_NUM_THREADS=4`
- **OMP_NUM_THREADS** is one of the environment variables defined in the standard
- If you don't specify the number of threads the default value is implementation defined (i.e. the standard doesn't say what it has to be)

Unit Summary

- Used the “Hello World” program from previous unit (`#pragma`) which contains an OpenMP **compiler directive**
- We need to use a compiler flag to enable OpenMP and this flag is compiler dependent
- We can control the number of threads forked in parallel region using the `OMP_NUM_THREADS` **environment variable**